THE LOCALLY RINGED ETALE TOPOS CONSTRUCTION IS FULLY FAITHFUL ON ALGEBRAIC SPACES OVER THE INTEGERS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original lemma 04M6: sources/stacks-chow/tags.txt, source lines 6032–6032. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 04M1: sources/stacks-chow/tags.txt, source lines 6027–6027. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04M5: sources/stacks-chow/tags.txt, source lines 6031–6031. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04J8: sources/stacks-chow/tags.txt, source lines 5929–5929. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04I7: sources/stacks-chow/tags.txt, source lines 5893–5893. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0262: sources/stacks-chow/tags.txt, source lines 2835–2835. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Theorem 0.1. *Let X, Y be algebraic spaces over* $\mathrm{Spec}(\mathbf{Z})$. *Let*

$$(g, g^\sharp) : (Sh(X_{étale}), \mathcal{O}_X) \longrightarrow (Sh(Y_{étale}), \mathcal{O}_Y)$$

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

be a morphism of locally ringed topoi. Then there exists a unique morphism of algebraic spaces $f : X \to Y$ such that $(g, g^\sharp)$ is isomorphic to $(f_{small}, f^\sharp)$. In other words, the construction

$$\textit{Spaces}/\operatorname{Spec}(\mathbf{Z}) \longrightarrow \textit{Locally ringed topoi}, \quad X \longrightarrow (X_{\acute{e}tale}, \mathcal{O}_X)$$

is fully faithful (morphisms up to 2-isomorphisms on the right hand side).

Proof. The uniqueness we have seen in Lemma 04M6. Thus it suffices to prove existence. In this proof we will freely use the identifications of Equation (04M1) as well as the result of Lemma 04M5.

Let $U \in \mathrm{Ob}(X_{\acute{e}tale})$, let $V \in \mathrm{Ob}(Y_{\acute{e}tale})$ and let $s \in g^{-1}h_V(U)$ be a section. We may think of s as a map of sheaves $s : h_U \to g^{-1}h_V$. By Modules on Sites, Lemma 04J8 we obtain a commutative diagram of morphisms of ringed topoi

$$\begin{array}{ccc} (Sh(X_{\acute{e}tale}/U), \mathcal{O}_U) & \xrightarrow[(j,j^\sharp)]{} & (Sh(X_{\acute{e}tale}), \mathcal{O}_X) \\ \downarrow{\scriptstyle (g_s, g_s^\sharp)} & & \downarrow{\scriptstyle (g,g^\sharp)} \\ (Sh(V_{\acute{e}tale}), \mathcal{O}_V) & \longrightarrow & (Sh(Y_{\acute{e}tale}), \mathcal{O}_Y). \end{array}$$

By Étale Cohomology, Theorem 04I7 we obtain a unique morphism of schemes $f_s : U \to V$ such that $(g_s, g_s^\sharp)$ is 2-isomorphic to $(f_{s,small}, f_s^\sharp)$. The construction $(U, V, s) \rightsquigarrow f_s$ just explained satisfies the following functoriality property: Suppose given morphisms $a : U' \to U$ in $X_{\acute{e}tale}$ and $b : V' \to V$ in $Y_{\acute{e}tale}$ and a map $s' : h_{U'} \to g^{-1}h_{V'}$ such that the diagram

$$\begin{array}{ccc} h_{U'} & \xrightarrow[s']{} & g^{-1}h_{V'} \\ \downarrow{\scriptstyle a} & & \downarrow{\scriptstyle g^{-1}b} \\ h_U & \xrightarrow{s} & g^{-1}h_V \end{array}$$

commutes. Then the diagram

$$\begin{array}{ccc} U' & \xrightarrow[f_{s'}]{} & u(V') \\ \downarrow{\scriptstyle a} & & \downarrow{\scriptstyle u(b)} \\ U & \xrightarrow{f_s} & u(V) \end{array}$$

of schemes commutes. The reason this is true is that the same condition holds for the morphisms $(g_s, g_s^\sharp)$ constructed in Modules on Sites, Lemma 04J8 and the uniqueness in Étale Cohomology, Theorem 04I7.

The problem is to glue the morphisms f_s to a morphism of algebraic spaces. To do this first choose a scheme V and a surjective étale morphism $V \to Y$. This means that $h_V \to *$ is surjective and hence $g^{-1}h_V \to *$ is surjective too. This means there exists a scheme U and a surjective étale morphism $U \to X$ and a morphism $s : h_U \to g^{-1}h_V$. Next, set $R = V \times_Y V$ and $R' = U \times_X U$. Then we get $g^{-1}h_R = g^{-1}h_V \times g^{-1}h_V$ as g^{-1} is exact. Thus s induces a morphism $s \times s : h_{R'} \to g^{-1}h_R$. Applying the constructions above we see that we get a commutative diagram of morphisms of schemes

$$\begin{array}{ccc} R' & \xrightarrow[f_{s\times s}]{} & R \\ \downarrow\downarrow & & \downarrow\downarrow \\ U & \xrightarrow{f_s} & V \end{array}$$

Since we have $X = U/R'$ and $Y = V/R$ (see Spaces, Lemma 0262) we conclude that this diagram defines a morphism of algebraic spaces $f : X \to Y$ fitting into

an obvious commutative diagram. Now we still have to show that $(f_{small}, f^\sharp)$ is 2-isomorphic to $(g, g^\sharp)$. Let $t_V : f_{s,small}^{-1} \to g_s^{-1}$ and $t_R : f_{s\times s,small}^{-1} \to g_{s\times s}^{-1}$ be the 2-isomorphisms which are given to us by the construction above. Let $\mathcal{G}$ be a sheaf on $Y_{étale}$. Then we see that t_V defines an isomorphism

$$f_{small}^{-1}\mathcal{G}|_{U_{étale}} = f_{s,small}^{-1}\mathcal{G}|_{V_{étale}} \xrightarrow{t_V} g_s^{-1}\mathcal{G}|_{V_{étale}} = g^{-1}\mathcal{G}|_{U_{étale}}.$$

Moreover, this isomorphism pulled back to R' via either projection $R' \to U$ is the isomorphism

$$f_{small}^{-1}\mathcal{G}|_{R'_{étale}} = f_{s\times s,small}^{-1}\mathcal{G}|_{R_{étale}} \xrightarrow{t_R} g_{s\times s}^{-1}\mathcal{G}|_{R_{étale}} = g^{-1}\mathcal{G}|_{R'_{étale}}.$$

Since $\{U \to X\}$ is a covering in the site $X_{spaces,étale}$ this means the first displayed isomorphism descends to an isomorphism $t : f_{small}^{-1}\mathcal{G} \to g^{-1}\mathcal{G}$ of sheaves (small detail omitted). The isomorphism is functorial in $\mathcal{G}$ since t_V and t_R are transformations of functors. Finally, t is compatible with $f^\sharp$ and $g^\sharp$ as t_V and t_R are (some details omitted). This finishes the proof of the theorem. □